

Title

KrishiMitra AI

An Intelligent Farming Companion Powered by Gemma 4

Subtitle

KrishiMitra AI is an AI-powered agricultural assistant that helps farmers diagnose crop diseases, receive personalized farming guidance, analyze crop images, and make informed decisions using the reasoning capabilities of Gemma 4.

Problem Statement

Agriculture remains the primary source of livelihood for millions of people, yet many farmers struggle to access reliable agricultural expertise when they need it most. Limited access to specialists, delayed diagnoses of crop diseases, inconsistent internet connectivity, and language barriers often result in reduced crop yield and financial losses.

Existing digital solutions are either too complex, require constant internet connectivity, or fail to provide contextual recommendations that farmers can easily understand and apply.

KrishiMitra AI addresses these challenges by providing an intelligent agricultural assistant capable of understanding images and natural language while delivering clear, practical recommendations.

Our Solution

KrishiMitra AI combines artificial intelligence with an intuitive web application to provide farmers with instant agricultural support.

The platform enables users to:

- Detect crop diseases from uploaded images
- Receive AI-generated treatment recommendations
- Ask farming-related questions in natural language
- Get fertilizer and irrigation guidance
- Access weather and farming insights
- Receive responses in a farmer-friendly conversational style

Instead of providing generic responses, the system generates personalized recommendations based on the user's query and uploaded images.

Features

1. Multilingual AI Farming Assistant (Powered by Gemma 4)

Farmers can interact with KrishiMitra AI using natural language. Gemma 4 answers questions related to crop cultivation, pest management, irrigation, fertilizers, harvesting, and general farming practices in simple, easy-to-understand language. The assistant also supports contextual follow-up conversations.

2.Voice-Based AI Conversations

Farmers can interact with KrishiMitra AI using voice instead of typing. Spoken queries are converted into text, processed by Gemma 4, and the AI response is delivered in a natural, conversational manner, making the platform more accessible for users who are less comfortable with typing.

3. AI Crop Disease Detection

Farmers can upload an image of a crop or leaf for AI-powered disease analysis. The system identifies possible diseases, explains symptoms, provides confidence-based insights, and recommends preventive measures and treatment options with the help of Gemma 4.

4. Smart Weather Advisory

Instead of displaying only weather forecasts, KrishiMitra combines real-time weather information with Gemma 4 to generate practical farming advice, such as irrigation planning, fertilizer timing, and precautions before rainfall or extreme weather.

5. Crop Guide

Farmers can explore crop-specific information including growing season, water requirements, temperature range, expected yield, soil requirements, and cultivation basics. The guide provides quick and easy reference information to help farmers make informed decisions throughout the crop lifecycle.

6. Intelligent Irrigation Planning

Based on crop type, growth stage, and weather conditions, the system recommends whether irrigation is required and provides general water management suggestions.

7. Fertilizer Recommendation Assistant

KrishiMitra generates personalized fertilizer recommendations based on crop type, farm size, and growth stage instead of relying on fixed or static templates.

8. Market Price Insights

Farmers can view market prices and trends for selected crops. Gemma explains the data in simple language to help farmers make better selling decisions.

9. Farmer-Friendly Dashboard

A clean, responsive dashboard provides quick access to weather, crop health, disease detection, AI chat, irrigation, fertilizer planning, crop guides, and market insights from one place.

10. Regional Language Support

The application is designed to support multiple Indian languages such as English, Hindi, and Bengali, making AI assistance accessible to a wider farming community.

Why We Chose Gemma 4

Gemma 4 serves as the intelligence engine behind KrishiMitra AI.

We selected Gemma 4 because it provides:

- Strong multimodal understanding for image-based crop analysis
- Excellent reasoning capabilities for agricultural problem solving
- High-quality instruction following
- Context-aware conversations
- Fast inference suitable for real-time assistance
- A deployment path that supports scalable and potentially offline agricultural applications

Rather than relying on fixed rules or predefined decision trees, Gemma 4 dynamically analyzes each user request and generates explanations tailored to the farmer's specific situation.

Technical Architecture

Frontend

- React
- TypeScript
- Vite
- Tailwind CSS
- Framer Motion

Backend

- Firebase
- Firestore Database

Artificial Intelligence

- Google GenAI SDK
- Gemma 4

Data Flow

User Input – React Frontend – Firebase Backend – Gemma 4 API –AI Processing –
Personalized Recommendation –Farmer Dashboard

Features

Crop Disease Detection

Farmers can upload crop images for AI-powered disease analysis. Gemma 4 identifies visible symptoms, explains possible diseases, and recommends treatment methods.

AI Farming Assistant

Users can ask agricultural questions naturally and receive detailed responses covering cultivation techniques, pest management, irrigation schedules, fertilizer recommendations, and best farming practices.

Personalized Recommendations

Recommendations are generated dynamically based on user inputs rather than selected from static templates.

User-Friendly Interface

The application features a responsive interface designed for accessibility across desktop and mobile devices.

Technical Highlights

- Gemma 4-powered multilingual conversational AI
- AI-assisted crop disease detection using image understanding
- Real-time weather integration for intelligent farming recommendations
- Responsive React frontend with modern UI
- Firebase cloud backend for authentication and data management
- Structured prompt engineering for agriculture-specific AI responses
- Modular and scalable architecture designed for future expansion

Impact

KrishiMitra AI has the potential to benefit:

- Small-scale farmers
- Agricultural students
- Rural communities
- Agricultural extension workers
- First-time farmers

By providing instant, AI-powered guidance, the platform can reduce dependency on delayed expert consultations while improving farming decisions.

Future Improvements

Future development will focus on:

- Offline AI support for low-connectivity rural areas
- Support for additional Indian regional languages
- Government agriculture scheme integration
- IoT sensor connectivity for smart farming
- Satellite-based crop health monitoring
- AI-powered crop yield prediction
- Farm expense tracking and analytics

Conclusion

KrishiMitra AI demonstrates how Gemma 4 can be applied to solve real-world agricultural challenges. By combining multimodal AI reasoning with an intuitive web interface, the platform delivers personalized farming assistance that is scalable, accessible, and practical for modern agriculture.

Proof of Work

1. Project Overview

KrishiMitra AI is an AI-powered agricultural assistant built during a 24-hour hackathon. The objective was to create a practical solution capable of helping farmers identify crop diseases, answer agricultural questions, and provide actionable recommendations using Gemma 4.

The project combines a modern React frontend, Firebase backend services, and Gemma 4 to create an end-to-end intelligent farming assistant.

2. System Architecture

User React Frontend Firebase Backend Google GenAI SDK

Gemma 4 AI Recommendation Engine Farmer Dashboard

3. Frontend Architecture

The frontend was developed using React with TypeScript to create a responsive and maintainable application.

Key technologies include:

- React
- TypeScript
- Tailwind CSS
- Framer Motion
- Vite

Responsibilities include:

- User interaction
- Image uploads
- AI chat interface
- Dashboard visualization
- Responsive UI rendering

4. Backend Architecture

Firebase provides the backend infrastructure responsible for:

- Cloud data storage
- Conversation history management
- Real-time database operations
- Cloud integration
- User authentication

This architecture enables scalable and reliable data management while minimizing backend complexity during the hackathon.

5. Gemma 4 Integration

Gemma 4 is the core reasoning engine of KrishiMitra AI.

Its responsibilities include:

- Understanding agricultural questions
- Analyzing crop images
- Identifying possible diseases

- Explaining symptoms
- Generating treatment recommendations
- Providing preventive farming advice

Instead of returning predefined answers, Gemma 4 reasons over each individual query to produce contextual and personalized responses.

6. AI PROCESSING PIPELINE :

User Uploads Image
Image Processing
Prompt Construction
Gemma 4 Analysis
Disease Identification
Treatment Recommendation
Preventive Advice
Response Display

7. Prompt Engineering

To improve consistency and reduce hallucinations, prompts were structured to guide Gemma 4 toward agricultural reasoning.

Example prompt:

You are an experienced agricultural expert.

Analyze the uploaded crop image.

Identify:

- Possible disease
- Confidence level
- Visible symptoms
- Treatment
- Preventive measures
- Farmer-friendly explanation

If uncertain, clearly mention the uncertainty instead of making assumptions.

This structured prompting significantly improved response quality during testing.

8. Challenges Faced

Challenge 1 : Limited Development Time

The project was completed within a 24-hour hackathon. Feature prioritization was essential to ensure that the core AI workflow was fully functional before adding additional enhancements.

Challenge 2 : Reliable AI Responses

Agricultural advice must be accurate and understandable. We refined prompts to encourage structured, transparent responses and to reduce unsupported conclusions.

Challenge 3 : Image Variability

Crop images differed in quality, lighting, and orientation. Prompt engineering was used to improve the model's ability to reason about imperfect inputs while acknowledging uncertainty when necessary.

Challenge 4 : User Experience

The interface was designed to remain simple and responsive despite integrating multiple AI-powered features, ensuring a smooth experience across different devices.

9. Engineering Decisions

Why React?

Fast development, reusable components, and excellent support for responsive interfaces.

Why Firebase?

Rapid backend development with managed cloud infrastructure and minimal operational overhead.

Why Tailwind CSS?

Efficient UI development with consistent styling during a time-constrained hackathon.

Why Gemma 4?

Gemma 4 offers strong multimodal reasoning, high-quality instruction following, contextual understanding, and scalable deployment options, making it well suited for agricultural assistance.

10. Testing

The application was manually tested using:

- Healthy crop images
- Diseased crop images
- Different lighting conditions
- Empty user inputs
- Long conversational prompts
- Invalid image uploads
- Responsive layouts across devices

Testing focused on ensuring that AI responses remained relevant, understandable, and useful.

11. Future Scope

Planned enhancements include:

- Offline AI support
- Regional language translation
- Government agriculture scheme integration
- Satellite imagery analysis
- IoT-based farm monitoring
- Crop yield prediction
- Personalized seasonal farming plans

12. Conclusion

KrishiMitra AI demonstrates how Gemma 4 can be leveraged to address real agricultural challenges through multimodal reasoning and conversational intelligence. Despite being developed within a 24-hour hackathon, the project establishes a strong technical foundation for a scalable AI-powered farming assistant that can evolve into a production-ready solution with future enhancements.